

MAUS User Guide

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Note: This is a placeholder (“dummy”) guide scaffold. Replace the bracketed [TODO] sections with project-specific content before publishing.

Methyl Assignments Using Satisfiability — a clean-room reimplementaion of the SAT-based methyl NMR assignment method of Nerli et al., *Nat. Commun.* 12:691 (2021).

1. Overview

MAUS assigns methyl NMR resonances by **subgraph isomorphism** rather than by scoring. Given a protein structure and real **peak lists** (HMQC + NOESY), it returns — for each peak — the *set* of methyls consistent with every hard constraint. It never emits a single guess unless the data force one, and it provably never excludes a valid assignment.

Concept	In this tool
Structure graph G	one node per methyl carbon; geminal / short / long edges
Data graph H nodes	HMQC peaks: (¹ H, ¹³ C) shift + residue type
Data graph H edges	NOESY cross peaks, endpoints matched to HMQC by frequency
Solver	python-sat (Glucose3), assumption-based enumeration
Output	per-peak option set (1 / 2–3 / >3 / unassigned)

2. Installation

```
python3 -m pip install python-sat
git clone https://github.com/deepnmr/maus.git
cd maus
```

Requirements:

- Python ≥ 3.8
 - python-sat (provides the Glucose3 SAT solver)
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3. Quick start

```
# 1. build peak lists from a BMRB shift file + the PDB
python make_peaklists.py examples/mbp/1ANF.pdb
                        examples/mbp/bmr7114_3.str

# 2. run MAUS on the HMQC + NOESY peak lists (--truth scores against
      the key)
python maus.py examples/mbp/1ANF.pdb examples/mbp/hmqc.tsv
                examples/mbp/noesy.tsv \
                --truth examples/mbp/hmqc_true.tsv --tol-h 0.02 --tol-c 0.2 --out
                mbp_options.tsv
```

Expected summary:

```
methyls(G nodes)=192  HMQC peaks=192  NOESY cross peaks=825
NOE match: firm=508 ambiguous(dropped)=317 unmatched=0
unique(1 option)      = 130/192
ambiguous(2-3 options)= 19/192
ambiguous(>3 options) = 43/192
unassigned            = 0/192
truth in option set   = 192/192 = 100.0%
```

4. Input formats

4.1 PDB structure

Standard ATOM records. Only these residue types are parsed by default: ALA ILE LEU MET THR VAL. Chain A (or blank) is used.

4.2 HMQC peak list (TSV) — data-graph nodes (input)

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type
P1      0.828      24.510      L
```

- label — anonymous peak id (P1, P2, ...); leaks nothing about the answer
- H_ppm, C_ppm — methyl (^1H , ^{13}C) chemical shifts
- res_type — one-letter code (A I L M T V), known from labeling

Ground truth lives in a **separate** file, never in the input:

```
label <TAB> H_ppm <TAB> C_ppm <TAB> res_type <TAB> True
P1      0.828      24.510      L              L7CD1
```

Pass it with `--truth` to score; omit it to run blind.

4.3 NOESY peak list (TSV) — data-graph edges

```
peak_id <TAB> H1 <TAB> C1 <TAB> H2 <TAB> C2 <TAB> mix
X1      0.828    24.51   0.712    23.10   short
```

Each row is a methyl-methyl cross peak. During the run both endpoints are matched back to HMQC peaks by frequency (within `--tol-h/--tol-c`); `mix` $\in$ {short, long} tags the mixing-time distance class.

4.4 Generating the peak lists

`make_peaklists.py` builds all three files from a PDB and a BMRB NMR-STAR shift file (`bmrXXXX_3.str`): `hmqc.tsv` (input), `hmqc_true.tsv` (truth key), and `noesy.tsv`. HMQC (^1H , ^{13}C) coordinates are the real BMRB methyl shifts; NOESY cross peaks are emitted for structurally close methyl pairs.

5. Command-line options

Option	Default	Meaning
<code>--short-cut</code>	6.0	structure short-range edge cutoff (Å)
<code>--long-cut</code>	10.0	structure long-range edge cutoff (Å)
<code>--tol-h</code>	0.02	^1H NOESY→HMQC match tolerance (ppm)
<code>--tol-c</code>	0.20	^{13}C NOESY→HMQC match tolerance (ppm)
<code>--labeling</code>	A;I;L;M;T;V	residue types present
<code>--out</code>	–	write per-peak options TSV

`make_peaklists.py` options: `--noe-short` (6.0), `--noe-long` (8.0), `--keep-k` (12 nearest NOE partners per methyl).

6. Interpreting output

The `--out` TSV columns:

Column	Meaning
<code>label</code>	input peak identifier (anonymous)
<code>res_type</code>	one-letter residue type
<code>n_options</code>	number of valid methyls
<code>options</code>	comma-separated methyl labels
<code>truth</code>	ground-truth label (blank if no <code>--truth</code>)
<code>truth_in_set</code>	1 if <code>truth</code> $\in$ <code>options</code> , else 0 (blank if no <code>--truth</code>)

- **1 option** → uniquely assigned.

- **2–3 options** → residual ambiguity, usually geminal/local symmetry.
 - **>3 options** → weak local NOE network.
 - **0 options** → over-constrained inputs (check cutoffs).
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7. On the bundled benchmark

MAUS sees only the two peak lists. HMQC (^1H , ^{13}C) coordinates are the real BMRB 7114 shifts, and each NOESY endpoint is resolved back to an HMQC peak **by frequency**, not by identity — so the assignment is not circular and shift degeneracy has real consequences.

- truth in option set = 100% is the MAUS **guarantee**: a valid assignment is provably never excluded (not a measurement — a property of the exact enumeration).
- The residual ambiguity is **real**. NOESY cross peaks whose endpoint matches more than one HMQC peak cannot be pinned to a definite methyl pair and are dropped, so shift-degenerate methyls keep several options. Tighter `--tol-h/--tol-c` recovers more unique calls (170/192 at $\pm 0.01/0.1$) as fewer NOEs are ambiguous — the honest resolution/degeneracy trade-off.

The NOESY cross peaks themselves are still *generated from the structure* (close methyl pairs), so this remains a controlled benchmark rather than a picked experimental NOESY. Swap in a real NOESY peak list to go fully experimental — the `maus.py` interface does not change.

8. Troubleshooting

Symptom	Likely cause	Fix
ModuleNotFoundError: pysat	solver not installed	<code>pip install python-sat</code>
many unassigned peaks	structure cutoffs too tight	<code>raise --long-cut</code>
most peaks ambiguous	too many NOEs dropped as ambiguous	<code>tighten --tol-h/--tol-c</code>
high unmatched NOE count	tolerance too tight vs shift precision	<code>loosen --tol-h/--tol-c</code>

9. References

- Nerli, De Paula, McShan & Sgourakis. *Backbone-independent NMR resonance assignments of methyl probes in large proteins*. **Nat. Commun.** 12:691 (2021). [doi:10.1038/s41467-021-20984-0](https://doi.org/10.1038/s41467-021-20984-0)
 - See `COMPARISON.md` for the head-to-head against MAGIC.
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[TODO] Add author, license, and contact sections before release.